AI and Math KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

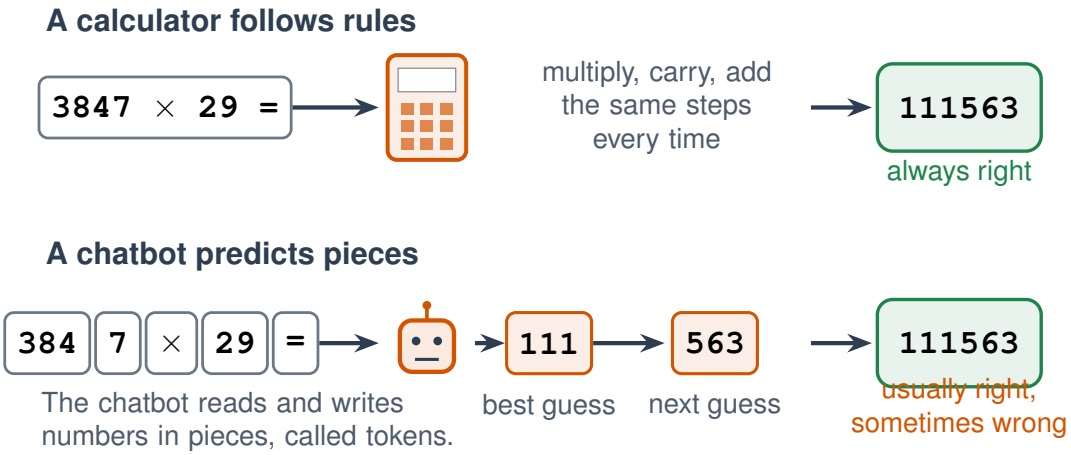
1. Catch the mistake. A chatbot solved $3x + 5 = 20$. Circle the wrong step.

Step	Work
1	$3x + 5 = 20$
2	$3x = 25$
3	$x = \frac{25}{3}$

The right answer: $x =$ 5

Check by plugging in: 3(5) + 5 = 20 ✓

2. Rules and predictions.



A calculator follows rules. A chatbot predicts the answer, one token at a time.

Why can a chatbot get a long problem wrong? One bad guess makes the whole answer wrong.

3. Three ways to check. Is $3847 \times 29 = 111563$?

Estimate	$4000 \times 30 = 120000$. Close!
Calculator	111563
Plug it in	$111563 \div 29 = 3847$

4. Test the chatbot. Ask for only the answer. Check it with a calculator.

Problem	The AI said	The calculator said	✓ or ✗
23×47		1081	
3847×29		111563	
48213×7956		383582628	
8374615×293847		2460855493905	

Which problems did the AI get wrong most often in our class? Answers vary. The biggest ones.

5. The hint challenge. Solve $2x - 7 = 11$. Do not let the AI give you the answer.

The AI's hint	My step
Undo the -7 first.	$2x = 18$
Now undo the $\times 2$.	$x = 9$

Check by plugging in: $2(9) - 7 = 11$ ✓

6. Talk about it.

AI helps me learn math when Answers vary. It gives hints and checks my steps.

AI stops me from learning when Answers vary. It does the problem for me.